

COMPUTER NETWORKS AND INTERNET ACCESS



LEAD-IN

Activity

- 1 What types of networks do you know?
- 2 Where are different types of networks used?
- 3 What do you need to connect to the Internet?
- 4 What types of connection do you know? How do they differ?



TOPIC

Text

The Internet is a global system of interconnected computer networks that use the standard Internet protocol suite – the **Transmission Control Protocol (TCP)** and the **Internet Protocol (IP)** to link several billion devices worldwide. It is an international network of networks that consists of millions of private, public, academic, business, and government networks, of local to global scope, that are linked by a broad array of electronic, wireless, and optical networking technologies. To communicate with one another computers need to use the same TCP/IP.

Computer Networks

There are the following types of computer networks:

Personal Area Network (PAN)

A PAN lets devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals.

Almost every computer has an attached monitor, keyboard, mouse, and printer. Without using wireless, this connection must be done with cables. So many new users have a hard time finding the right cables and plugging them into the right little holes. To help these users, some companies got together to design a short-range wireless network called **Bluetooth** to connect these components without wires. Bluetooth technology also enables wireless communication between laptop computers, mobile phones and PDAs. It enables devices use short-range radio signals to exchange data quickly and easily. It was developed by a group of computer and telecommunication companies including IBM, Intel, Nokia and Ericsson.

Local Area Network (LAN)

A LAN is two or more connected computers sharing certain resources in a relatively small geographic location, often in the same building. They can be built with two main types of architecture: **peer-to-peer**, where the two computers have the same capabilities, or **client-server**, where one computer acts as the server containing the main hard disk and controlling other **workstations** or **nodes**, all the devices linked in the network. Examples include home networks and office networks. A family of computer networking technologies for LANs is **Ethernet**.

The computers on the network can share data, and they can also access printers connected to the network. Computers in a LAN need to use the same **protocol**, or standard of communication. Ethernet is one of the most common protocols for LANs. A **router** is needed to link a LAN to another network.

To build a WLAN (Wireless LAN) you need **access points**, radio-based receiver-transmitters that are connected to the wired LAN, and **wireless adapters** installed in your computer to link it to the network. **Hotspots** are WLANs available for public use in places like airports, and hotels, but sometimes the service is also available outdoors (e.g. university campuses, squares, etc.).

Metropolitan Area Network (MAN)

A metropolitan area network (MAN) is larger than a LAN, covering an area of a few city blocks to the area of an entire city, often including the surrounding areas. A family of computer networking technologies for MANs is **Ethernet**.

Wide Area Network (WAN)

A WAN typically consists of two or more LANs. The computers are farther apart and are linked by telephone lines, dedicated telephone lines, or radio waves. The Internet is the largest Wide Area Network (WAN) in existence. The main transmission paths within a WAN are high-speed lines called **backbones**.

If a personal computer is connected to a network, it is called a **network workstation** (Note: This is different from the usage of the term workstation as a high-end microcomputer). If your PC is not connected to a network, it is referred to as a **standalone computer**. A **server** is a computer that 'serves' many different computers in a network by running specialized software and storing information. For example, webpages are stored on servers.

Network topology refers to the shape of a network. There are three basic physical topologies.

- **Star** – there is a central device to which all the workstations are directly connected. This central position can be occupied by a server, or a **hub** or **switch**, a connection point of the elements of a network that redistributes the data.
- **Bus** or **Line** – every workstation is connected to a main cable called a **bus**.
- **Ring** – the workstations are connected to one another in a closed loop configuration. There is no central hub that holds all the data, and communication is sent in one direction around the ring through the use of a **token** (a special series of bits that travels around the ring network).
- **Tree** – a group of stars connected to a central bus.

Internet Access

Internet access connects individual computer terminals, computers, mobile devices, and computer networks to the Internet, enabling users to access Internet services. Before you can access the Internet, there are three things you need: an **Internet service provider**, a **modem**, and a **web browser**.

An **Internet service provider (ISP)** is an organization that provides services for accessing, using, or participating in the Internet. Internet service providers may be organized in various forms, such as commercial, community-owned, non-profit, or privately owned. They employ a range of technologies to connect users to their network, such as **dial-up**, **broadband**, **3G** and **4G**.

- **Dial-up**
A dial-up connection (referred to as a **narrowband** connection) connects a computer to the Internet via a modem and a public telephone network. It is the cheapest option, but slower than other kinds of Internet access (the maximum theoretical connection speed is 56 Kbps). The user cannot use the Internet and the phone at the same time unless there are multiple phone lines. To communicate over a phone line, the computer needs a **modem**, which converts the **digital signals** from the computer into **analogue signals** so that data can be transmitted across the phone or cable network.
- **Broadband**
A broadband connection allows a very fast connection to the Internet. There are several ways of getting online with broadband – **ADSL**, **cable TV**, and **satellite**. Each of these types of access involves connecting to an access point using either a wired Ethernet connection or a wireless connection.

An **ISDN (Integrated Services Digital Network)** capable phone line can provide connection speeds of up to 64 Kbps (single channel) or 128 Kbps (dual channel) using a terminal adapter often referred to as an ISDN modem. The terminal adaptor removes the need to convert digital signals to analogue signals before they're sent down the telephone line, this results in a more reliable Internet connection.

An **ADSL (Asymmetric Digital Subscriber Line) connection** works over the telephone line and requires a special **ADSL modem**. An ADSL is much faster than a dial-up access (up to 24 Mbps). It does not require the phone service; therefore, the Internet and the phone can be used at the same time.

A **cable connection** (requiring a special cable modem) uses a cable TV line to transmit and receive data and requires a **cable modem**. It can be faster (up to 50 Mbps) than a dial-up access, but it is only available where a cable is available.

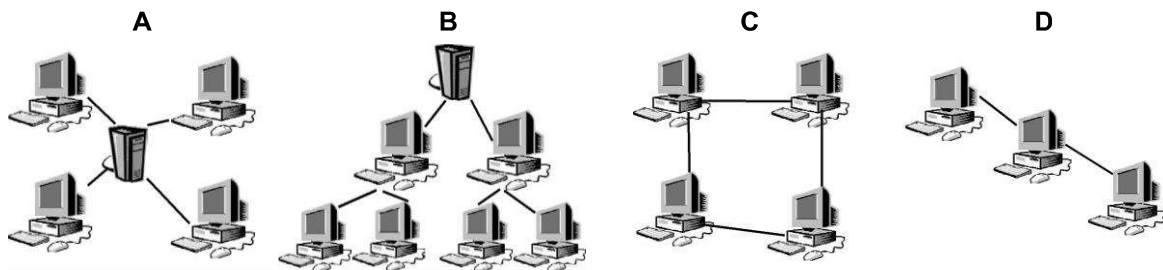
A **satellite connection** is provided through satellites and requires a **satellite modem** (it does not require phone or cable lines). This kind of access can be used almost anywhere in the world. It is faster than a dial-up access, but data is delayed, and its speed can be affected by the weather conditions.
- **3G and 4G**
3G and 4G (the **third and fourth generation** of mobile telecommunications technology) **systems** provide high-speed wireless Internet access even when there is no Wi-Fi connection and they are available for mobile phones and computers.

Task 1: Decide whether the following statements are true (T) or false (F).

- 1 LANs link computers and other devices that are placed far apart.
- 2 PANs can cover the area of an entire city.
- 3 The word 'protocol' refers to the shape of the network.
- 4 In client-server architecture the two computers have the same capabilities.
- 5 Access points don't need to be connected to a wired LAN.
- 6 Routers are used to link a LAN to another network.
- 7 Wireless adapters are optional when you are using a WLAN.
- 8 Hotspots can be found inside and outside buildings.
- 9 The Internet is an example of WAN.
- 10 Wireless WANs use fibre and cable as linking devices.
- 11 A standalone computer is not connected to a network.
- 12 A hub is in the central position in the ring topology.
- 13 A backbone is a high-speed line that creates the main transmission path within WAN.
- 14 Computers need to use the same protocol (TCP/IP) to communicate with each other.
- 15 ADSL and cable are two types of dial-up connections.
- 16 An ADSL connection is faster than a dial-up connection.
- 17 A cable connection uses existing phone lines and allows simultaneous Internet and phone usage.

Task 2: Identify the following network topologies.

- 1 line/bus topology
- 2 ring topology
- 3 star topology
- 4 hierarchical topology

**Task 3: Fill in the type of physical topology of communication networks.**

- 1: all the devices are connected to a central station.
- 2: in this type of network there is a cable to which all the computers and peripherals are connected.
- 3: two or more star networks connected together; the central computers are connected to a main bus.
- 4: all devices (computers, printers, etc.) are connected to one another forming a continuous loop.

**Speaking 1**

Task 1: In pairs, describe how different network topologies work.

Task 2: In groups, discuss and write down a list of the advantages and disadvantages of using different network topologies.



Listening 1

Listen to a description of how a web of networks works. For questions 1 – 10, choose the correct answer (A, B or C) which fits best according to what you hear.

- 1 We say that networks are alive because
 - A it requires a human to understand their concept.
 - B they still evolve.
 - C they form webs that make them look like a living organism.
- 2 A power function concept is NOT described as
 - A the product of a constant and a power of the independent variable.
 - B a big change in one parameter caused by a small change in another parameter.
 - C a function in which an independent variable appears in one of the exponents.
- 3 According to Google's algorithm, number of nodes with the number of references to a search object.
 - A increases
 - B corresponds
 - C links
- 4 As a node increases in size, it eventually becomes
 - A hub.
 - B centre.
 - C network.
- 5 Networks are so powerful because
 - A they are archives of historical information that apply to every person that uses the Internet.
 - B they are the only means to investigate events and activities surrounding major media events.
 - C as daily searches are collected, new clusters of links will rapidly form creating more networks.
- 6 The law that applies to our social networks is
 - A Kleinberg's Law of Social Interaction.
 - B Law of Six Degrees of Separation.
 - C Law of Logarithmic Expansion.



Reading

Networking Hardware

Connecting to the Internet requires certain networking hardware, including a **modem**, **router** or **network interface card** and an **access point**, providing a connection. The speed of an Internet connection is known as its (1).

Modem

A modem (which stands for **modulator-demodulator**) is a device that **modulates** the **digital signal** from the computer into **analogue signal** that can be carried over the phone line. The modem also (2) the analogue signal from the phone line into the digital signal, which it passes to the computer.

Router

A router is a specialised computer connected to at least two networks, commonly two LANs or WANs or a LAN and its ISPs network. It forwards data packets and allows you to share a single IP address among multiple network clients. Many routers are (3), allowing you to easily create a **wireless network**.

Network interface card

A network interface card (**NIC**, also known as a **network interface controller**, **network adapter**, **LAN adapter**) is a computer hardware component that connects a computer to a computer network. The network card can have a/an (4), a wireless connection, or both. The low cost and ubiquity of the Ethernet standard means that the newest computers have a network interface built into the motherboard. NIC-based connections are widely used in LANs, such as groups of computers in businesses. They can be used in homes, but many users prefer to use wireless connections for the added mobility.

Access points

Computers using a/an (5) and **Ethernet cable** connect through an access point. Access points are generally (6), **cable modems**, or **ADSL modems** that provide a link between the Internet service provider and the physical computer. They must be connected to a wired LAN. A wireless access point allows you to connect to an access point without using a physical connection. A wireless access can be configured in your home using a **wireless router** and a computer with a/an (7). Many businesses, such as hotels and coffee shops, provide free wireless access in their buildings for the use of their customers.

Task 1: Fill the gaps 1 – 7 with the missing words A – G.

- A routers
- B demodulates
- C Ethernet port
- D wireless interface
- E bandwidth
- F wireless
- G NIC

Task 2: Choose the correct answer.

- | | |
|---|--|
| <p>1 The speed of the Internet connection is known as its</p> <ul style="list-style-type: none"> A bandwidth B ISP C access point D broadband | <p>4 NIC stands for</p> <ul style="list-style-type: none"> A Network Internet Card B Network Interface Controller C Network Interceptor Card D Network Integrated Card |
| <p>2 A device that converts digital computer data to analogue signals is</p> <ul style="list-style-type: none"> A web browser B Ethernet C modem D I/O port | <p>5 A common networking technology is</p> <ul style="list-style-type: none"> A web browser B Ethernet C modem D I/O port |
| <p>3 A router is</p> <ul style="list-style-type: none"> A a type of circuit board inside all modems B a specialized computer-based device C a useful concept for understanding all modems D none of the above | <p>6 An access point is</p> <ul style="list-style-type: none"> A a router B a cable modem C an ADSL modem D all of the above |



Vocabulary Practice

Task 1: Complete the text on LAN using the words below:

LAN is pronounced 'lan', and stands for (1) Area Network. In a typical LAN, there is a central network (2) which supports a number of (3). Users have to (4) the network server. Pages of information that can be viewed within a LAN are called an (5). A number of LANs connected to each other via (6) or other form of communication are called a (7). To be used as network terminals, each computer needs to have a (8) installed.

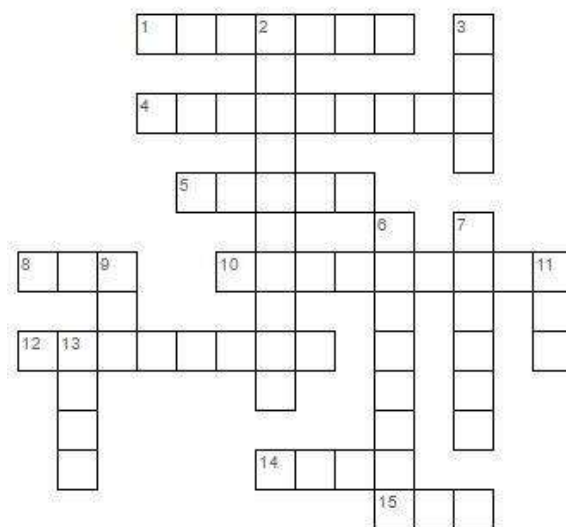
A intranet
B Local

C log onto
D network card

E satellite
F server

G terminals
H WAN

Task 2: Complete the crossword puzzle.



Across

- 1** a specific geographic location in which an access point provides public wireless broadband network services to mobile visitors through a WLAN
- 4** a type of data transmission in which a single medium (coaxial cable or optical fibre) can carry several channels at once
- 5** a device that modulates one or more carrier wave signals to encode digital information for transmission and demodulates signals to decode the transmitted information
- 8** a computer network that lets devices communicate over the range of a person
- 10** a wireless technology standard for exchanging data over short distances from fixed and mobile devices, and building PANs
- 12** the main transmission path within a WAN
- 14** a connection point; a device linked in a network
- 15** enables two hosts to establish a connection and exchange streams of data

Down

- 2** refers to a computer that is not connected to a network
- 3** an international communications standard for sending voice, video, and data over digital telephone lines or normal telephone wires.
- 6** a family of computer networking technologies commonly used in LANs and MANs
- 7** a device that forwards data packets along two or more networks
- 9** a computer hardware component that connects a computer to a computer network
- 11** a connection point of the elements of a star network topology
- 13** a type of broadband communications technology which is up to 24 Mbps fast



Language Functions: Troubleshooting

Troubleshooting is **the process of identifying and correcting faults in a computer system**. A service that gives people information and help, especially if they are having problems with a computer, is called a **help desk**. **Help desk technicians** offer computer assistance for all types of problems. They try to **identify their source** and to **propose a solution to them**.

Identifying the source of a problem:

Questions

What's gone wrong?
What's up?
Any idea what the problem is?
What do you have trouble with?

Answers

We're/I'm in big trouble.
I've got a problem... that's why I'm...
It looks like it's the.../something to do with...
I have trouble with my...

Proposing a solution:

Questions

How do you suggest I deal with this?
What's the best way to fix this?
How can I sort this out?

Answers

Let's just take it one step at a time.
The solution to this problem will require...
You'll just have to...
You should.../You'd better...
Why don't you...?

Keeping track:

Questions

Is everything working/going smoothly?
Can you just confirm/check that you...?
So, how did it go?

Answers

Yeah. It's working just as I wanted.
Yes, I've checked that .
It couldn't be better.
Not that well, actually.

Match the pieces of advice A – I with the problems 1 – 9.

- | | |
|---|---|
| <p>1 My printer is producing fuzzy, not clear, printouts.
 2 I get a lot of error messages. Some of my files won't open. They're corrupted, damaged.
 3 The monitor flickers, the image is unsteady.
 4 My optical drive fails, it won't read or write discs.
 5 My machine is running very slowly, and it shows low memory error messages.
 6 My computer is behaving strangely. I think it's got a virus.
 7 I get a 401 message: I'm unauthorized, not allowed to enter that website.
 8 I've tried to access a website, but I get a 404 Not Found message, as if it didn't exist.
 9 I try to connect but I get this message: Network connection refused by server.</p> | <p>A Haven't you got any antivirus software installed? If I were you, I'd try a free online scan.
 B Why don't you reset the refresh rate of your monitor?
 C You may have made a mistake while typing your password. You can't access a website if you aren't recognized as a guest. Try typing again.
 D The computer may be overheating. Check there's nothing blocking the flow of air. You should also use a recovery tool to retrieve your files.
 E Your system must be running short of memory. You'd better add some more RAM.
 F That message shows the web server is busy. Why don't you wait and try again later?
 G Your discs or perhaps your lens might be dirty. Use some special disc polish.
 H It might mean that the page is no longer on the internet. Check the URI again. If that doesn't work, you could use a search engine to find similar pages.
 I The print heads of your printer must be clogged, obstructed with ink. Run the clean cartridge routine or wipe them with a cloth and distilled water.</p> |
|---|---|